

Answers written in the margins will not be marked.

- (i) Draw an arrow in the figure below to represent the direction of the acceleration of the block at that moment. (1 mark)



- (ii) Find the horizontal displacement of the block from that moment until just before it reaches the ground. (3 marks)

- (iii) Describe the energy conversion of the block from that moment until just before it reaches the ground. (2 marks)

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(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

* (ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11} . Find its activity (in Bq) at $t = 0$ s. (2 marks)

* (iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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Figure 12.2(a)

Figure 12.2(b)

- (i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

- (ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

- (iii) Which type(s) of radiation (α , β or γ) can reach the chip of the ring ? (1 mark)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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